



Emmanuel Iarussi

Computer Vision | Computer Graphics | Machine Learning

Personal

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Interest Areas

AI ★★★★★
3D CG ★★★★★
Generative ★★★★★
Imaging ★★★★★
Vision ★★★★★

Languages

Spanish ★★★★★
English ★★★★★
French ★★★★★
Italian ★★★★★

Since 2017	Research Fellow CONICET	Buenos Aires, Argentina
Since 2022	Professor - Tenure Track Universidad Torcuato Di Tella	Buenos Aires, Argentina
2016 - 2017	Postdoctoral Fellow ISTA Austria - Digital Fabrication	Vienna, Austria
2012 - 2015	PhD. Computer Science INRIA GraphDeco Team - <i>Link to PhD Thesis</i> Advisors: Adrien Bousseau, George Drettakis	Sophia Antipolis, France
2006 - 2012	Systems Engineer & Programmer UNICEN - Facultad de Ciencias Exactas	Tandil, Argentina

Selected Publications

VesselGPT: Autoregressive Modeling of Vascular Geometry (Paper Highlight) Feldman, P., Sinnona, M., Delrieux, C. Siless, V., **Iarussi, E.** *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI 2025)*. Cham: Springer Nature Switzerland, 2025. doi.org/10.1007/978-3-032-05325-1_63

Recursive Variational Autoencoders for 3D Blood Vessel Generative Modeling. Feldman, P., Fainstein, M., Siless, V., Delrieux, C., **Iarussi, E.** *Medical Image Analysis*, 103703 (2025). doi: 10.1016/j.media.2025.103703

Enhancing and advancements in deep learning for melanoma detection: A comprehensive review. Virgens, G.S., **Iarussi, E.**, et al. *Computers in Biology and Medicine* (2025): 110533. doi:doi.org/10.1016/j.compbio.2025.110533

Improving Realism in Abdominal Ultrasound Simulation Combining a Segmentation-Guided Loss and Polar Coordinates training. Vitale, S., **Iarussi, E.**, Orlando, J.I., and Larrabide, I. *Medical Physics* (2025). doi:doi.org/10.1002/mp.17801

DUDF: Differentiable Unsigned Distance Fields with Hyperbolic Scaling. Fainstein, M., Siless, V., & **Iarussi, E.** In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2024)*.

VesselVAE: Recursive Variational Autoencoders for 3D Blood Vessel Synthesis. Feldman, P., Fainstein, M., Siless, V., Delrieux, C., **Iarussi, E.** *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI 2023)*. *Lecture Notes in Computer Science*, vol 14220. Springer, Cham. doi:10.1007/978-3-031-43907-0_7.

Bone-GAN: Generation of virtual bone microstructure of high resolution peripheral quantitative computed tomography. Thomsen, F. S. L., **Iarussi, E.**, Borggreffe, J., Boyd, S. K., Wang, Y., Battié, M. C. *Medical Physics* (2023). ISSN: 2473-4209. doi: 02/mp.16482.

Learning normal asymmetry representations for homologous brain structures. Deangeli, D., **Iarussi, E.**, Princich, J. P., Bendersky, M., Larrabide, I., Orlando, J. *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI 2023)*. *Lecture Notes in Computer Science*, vol 14220. Springer, Cham. doi: 10.1007/978-3-031-43993-3_8.

NORHA: A NORmal Hippocampal Asymmetry deviation index based on one-class novelty detection and 3D shape features. Deangeli, D., Iarussi, F., Kølsgaard, H., Braggio, D., Princich, J. P., Bendersky, M., **Iarussi, E.**, Larrabide, I., Orlando, J. I. *Brain Topography* (2023). ISSN: 1573-6792. doi:s10548-023-00985-6. **Learning Deep Features for**

Stain-free Live-dead Human Breast Cancer Cell Classification. Pattarone, G., Acion, L., Simian, M., **Iarussi, E.** *Nature Scientific Reports* (2021). ISSN: 2045-2322. doi: 10.1038/s41598-021-89895-w.

Generative Modelling of 3D in-silico Spongiosa with Controllable Micro-Structural Parameters. Iarussi, E., Thomsen, F. and Delrieux, C., *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI 2020)*. *Lecture Notes in Computer Science*, vol 12266. Springer, Cham. doi: 10.1007/978-3-030-59725-2_76.

Improving realism in patient-specific abdominal Ultrasound simulation using CycleGANs. Vitale, S., Orlando, J.I., Iarussi, E. and Larrabide, I. *International Journal of Computer Assisted Radiology and Surgery*, 1-10 (2020). ISSN: 1861-6429. doi: 10.1007/s11548-019-02046-5.

FlexMaps: Computational Design of Flat Flexible Shells for Shaping 3D Objects. Malomo, L., Pérez, J., Iarussi, E., Pietroni, N., Miguel, E., Cignoni, P. and Bickel, B. *ACM Transactions on Graphics (SIGGRAPH Asia)* 37.6 (2018). ISSN: 0730-0301. doi: 10.1145/3272127.3275076.

CoreCavity: Interactive Shell Decomposition for Fabrication with Two-Piece Rigid Molds. Kazutaka Nakashima, Thomas Auzinger, Emmanuel Iarussi, Ran Zhang, Takeo Igarashi, Bernd Bickel. *ACM Transactions on Graphics (SIGGRAPH)* 37.4 (2018). ISSN: 0730-0301. doi: 10.1145/3197517.3201341..

WrapIt: Computer-Assisted Crafting of Wire Wrapped Jewelry. Emmanuel Iarussi, Wilmot Li, and Adrien Bousseau. *ACM Transactions on Graphics (SIGGRAPH Asia)* 34.6 (2015). doi:10.1145/2816795.2818118. ISSN: 0730-0301.

BendFields: Regularized Curvature Fields from Rough Concept Sketches. Emmanuel Iarussi, David Bommes, and Adrien Bousseau. *ACM Transactions on Graphics (TOG)* 34.3 (2015): 24. ISSN: 0730-0301. doi:10.1145/2710026.

The Drawing Assistant: Automated Drawing Guidance and Feedback from Photographs. Emmanuel Iarussi, Adrien Bousseau, and Theophanis Tsandilas. *ACM Symposium on User Interface Software and Technology (UIST)*. ACM, 2013. ISBN: 978-1-4503-8635-7. doi:10.1145/2501988.2501997

Awards

2024 — Second Best Paper Award (LatinX - CVPR 2024).

2022 — Top 100 cancer Nature Scientific Reports papers in 2021.

2021 — Salesforce AI Research Grant. Grant: USD 50,000.00.

2020 — Kaggle Open Data Research Grant 2020. Grant: USD 2,500.00.

Research Funding

2024 — SLOAN Foundation. *VisDecode: AI system for automated interpretation and feedback on scientific plots*. Grant: USD 70,000.00.

2023 — Project CLIAS – IECS. *Remote AI system for diagnosis of cervicovaginal smear images (PAPs)*. Grant: USD 40,000.00.

2023 — Civic Health Project. *Tools to reduce polarization in Telegram using LLMs*. Grant: USD 5,000.00.

2021 — PIP / ID: PIP 2021–2023 GI – 11220200102981CO. *AI models for osteoporosis diagnosis from routine CT images*. Grant: USD 10,000.00.

2020 — PID UTN / ID: SIUTNBA0005534. *Generative networks for interactive 2D/3D multi-view synthesis*. Grant: \$644,225.68.

2020 — PID UTN / ID: SIUTNBA0005139. *CrossMatch: Deep learning for cross-domain correspondence detection*. Grant: \$1,143,878.00.

2019 — PICT-Joven / ID: PICT-2018-04517. *Deep multi-view descriptors for line drawing matching*. Grant: USD 3,500.00.

2017 — Nvidia Research Grant. *Cross-domain feature learning for 2D/3D matching with CNNs*. Grant: USD 3,800.00.